Multiscale QALT

Exact Conditional Fibers with Latent-Order Iterative Work

Theory and registered empirical audit draft

23 August 2026

Abstract

Full-dimensional diffusion preserves endpoint information but repeatedly processes every image or video coefficient. Latent diffusion reduces that work through a learned tokenizer whose decoder can lose density or impose a restricted conditional family. Multiscale QALT instead applies a fixed local orthonormal chart, iteratively generates only the coarse field, and samples every detail coefficient once from a parent-conditional fiber. On a declared multiscale law, the construction exactly matches the full endpoint density with work $O(Kn_a + N)$, the same order as latent diffusion, while retaining all N transformed coefficients. A parent-gated Gaussian-mixture fiber has a strict log-score advantage over a fixed diagonal Gaussian decoder at the same coarse dimension when the mixture is not Gaussian. The result is conditional: an optimized same-information latent baseline can use the identical fiber and tie both quality and order of work. The registered image/video assay therefore tests finite recipes and restricted decoder families rather than universal dominance.

1 Problem and construction

Let $x \in \mathbb{R}^N$ denote an image or video, and let W be a one-level separable orthonormal Haar transform implemented by local pairwise lifting. Write $Wx = (a, h)$, where $a \in \mathbb{R}^{n_a}$ is the all-low-pass field and $h \in \mathbb{R}^{N-n_a}$ contains every remaining band. The representation is equal-dimensional because W is bijective; only the state receiving repeated neural updates has dimension n_a.

The declared conditional law factors as

$$p(x \mid c) = p_a(a \mid c) \prod_{\ell,i} p_\ell\big(h_{\ell i} \mid u_{\ell i}, c\big), \qquad (a, h) = Wx, \tag{1}$$

where $u_{\ell i}$ is a local parent formed from coarse or earlier-scale coefficients. The executable first uses the nonlinear non-Gaussian fiber

$$p_\ell(h \mid u) = \{1 - \pi_\ell(u)\}\mathcal{N}(0, s_{\ell 0}^2) + \pi_\ell(u)\mathcal{N}(0, s_{\ell 1}^2), \quad \pi_\ell(u) = \operatorname{logit}^{-1}\{\beta_\ell^{\mathsf{T}}(1, \tanh u, \sin u)\}. \tag{2}$$

All mixture parameters are fitted from training coefficients by expectation–maximization. No component label or population parameter enters fitting.

2 Endpoint law and density

Theorem 2.1 (Exact endpoint and likelihood)**.** *Assume that the active sampler returns $a \sim p_a(\cdot \mid c)$ and that each one-shot fiber sampler returns the conditional distribution in Equation* (1) *in an order* *consistent with its parents. Then* $x = W^\top(a, h)$ *has density* $p(x \mid c)$*. Moreover,*

$$\log p(x \mid c) = \log p_a(a \mid c) + \sum_{\ell,i} \log p_\ell(h_{\ell i} \mid u_{\ell i}, c), \tag{3}$$

with no chart log-Jacobian term.

Proof. Sequential conditional sampling constructs the joint density on the right side of Equation (1). Orthonormality gives $W^{-1} = W^\top$ and $|\det W| = 1$. The change-of-variables formula therefore preserves the density value and yields Equation (3). □

This theorem gives equality with a realizable full-dimensional flow or diffusion at its exact endpoint. It does not assert that arbitrary observed data satisfy the factorization or that a fitted finite model realizes every conditional.

3 Work, memory, and optimized controls

Let K be the number of repeated sampler evaluations. Count one local token update as constant work after fixing channel width and local receptive field.

Proposition 3.1 (Conditional complexity parity)**.** *A full-token local sampler uses* $\Theta(KN)$ *token updates and* $\Omega(N)$ *state memory. Multiscale QALT uses*

$$Kn_a + (N - n_a) \tag{4}$$

token updates, plus $\Theta(N)$ *local chart work and* $\Theta(N)$ *output memory. Thus its total work is* $\Theta(Kn_a + N)$*, the same asymptotic order as a latent sampler with an* n_a*-token state and a linear-time decoder. For* $K > 1$ *and* $n_a < N$*, Equation* (4) *is strictly smaller than* KN*.*

Proof. The active state is updated K times. Each detail is sampled once, and the lifting chart touches each coefficient a constant number of times. A generated endpoint itself requires $\Omega(N)$ memory. Summing these terms proves the result. □

Constants remain empirical. Dense attention would change the count, and few-step distillation can reduce K. The registered study consequently reports latency, live working arrays, token updates, and sampler evaluations rather than inferring hardware speed from asymptotic notation alone.

Theorem 3.2 (Same-information tie)**.** *Consider a latent baseline with the same chart, active sampler, local parents, fitted fibers, base random variables, and exact one-shot conditional sampler. It induces exactly the same distribution and has the same* $\Theta(Kn_a + N)$ *work as Multiscale QALT.*

Proof. The two algorithms are identical measurable maps of identically distributed base variables. Their endpoint laws and operation counts coincide. □

The tie rules out universal quality or efficiency superiority over unrestricted latent diffusion. A strict empirical claim must name the finite solver, lossy tokenizer, decoder restriction, or architecture that creates the gap.

4 Restricted quality separations

Proposition 4.1 (Fixed diagonal-decoder gap)**.** *Fix the chart, coarse representation, active law, conditioning, and conditional mean. Let the true fiber be Equation* (2) *with* $0 < \pi_\ell(u) < 1$ *on a set of positive probability and* $s_{\ell 0} \neq s_{\ell 1}$*. For any fixed conditional diagonal Gaussian decoder* $q(h \mid a, c)$*,*

$$\mathbb{E}_{a,c}\, \mathrm{KL}\{p(h \mid a, c) \,\|\, q(h \mid a, c)\} > 0, \tag{5}$$

unless the decoder family contains the true conditional fiber. The excess expected negative log likelihood equals the left side of Equation (5)*.*

Proof. The excess log score is the conditional forward Kullback–Leibler divergence. A nondegenerate centered mixture with unequal component variances is not Gaussian, as follows, for example, from its fourth cumulant. Gibbs' inequality is strict whenever the two conditional densities differ on a set of positive measure. Summing conditionally independent detail terms preserves strict positivity. □

This is a representation result only relative to the fixed chart, equal coarse dimension, and declared decoder family. It is not dominance over an unrestricted variational autoencoder, whose decoder could implement the same mixture.

For a finite-Euler control, transporting a standard normal scale s through $r'(t) = (\log s)r(t)$ for K steps gives scale

$$s_K = \{1 + (\log s)/K\}^K. \tag{6}$$

Whenever $s \neq 1$ and the stability factor is positive, $s_K \neq s$. Applying this replacement to both identifiable unequal-scale mixture components changes the conditional distribution; hence its forward Kullback–Leibler divergence from the exact fiber is strictly positive. An exponential or structural split integrator uses the exact scale and ties QALT.

5 Registered evidence ladder

Stage A uses procedurally generated 32×32 image-shaped arrays and $16 \times 32 \times 32$ video-shaped arrays. Coarse coefficients have a nonlinear heavy-tailed mixture law, and detail coefficients follow Equation (2). Development seeds are 700–704; confirmation seeds 800–829 remain inaccessible until a separate freeze commit. Every method receives identical observations, conditions, coefficient dimension, local parents, and base randomness. Training alone fits fiber parameters; the active law is sampled from the procedural truth. Validation is diagnostic; test data are read once for frozen metrics.

Mandatory controls are the oracle law, full-token exact and Euler samplers, a coarse-only decoder, Multiscale QALT, the exact split tie, and the fixed diagonal Gaussian decoder. Passing Stage A would establish the theorem-to-implementation mechanism on the declared law. It would not establish observed image or video superiority; that requires a separately frozen study against optimized latent, wavelet, and few-step baselines.

Table 1: The corrected development assay passes every frozen gate on five seeds. Differences are nats per full endpoint dimension; ratios include the matched local update kernel and shared inverse Haar decoder. Intervals are paired 95% Student intervals.

Quantity	Image	Video
Fitted − oracle conditional detail NLL	0.000015	0.000044
Diagonal − mixture detail NLL	0.1030	0.1073
Unit-Gaussian − mixture detail NLL	0.1074	0.1156
Finite Euler − exact detail NLL	0.000204	0.000129
Local-kernel timing proxy ratio	0.328	0.192
Formula live-array ratio	0.750	0.625

The exact full-token and same-information split controls tie QALT to numerical precision. Thus the table supports the declared mechanism and restricted separations, not universal quality dominance. The memory row counts explicit simultaneously live arrays rather than accelerator allocator telemetry, and the procedural endpoint law is not observed-data evidence.

One subsequent confirmation used the frozen implementation and untouched seeds 800–829. All 16 paired one-sided Student components passed Holm correction at familywise level 0.05, together with every deterministic hard check. Fitted-minus-oracle conditional detail NLL, normalized by full endpoint dimension, was 0.000121 for image-shaped arrays and 0.000080 for video-shaped arrays. Diagonal-minus-mixture detail gaps were 0.10163 and 0.10612, while unit-Gaussian-detail gaps were 0.10541 and 0.11406. Proxy local-kernel timing ratios were 0.3274 and 0.1894; formula live-array ratios remained 0.750 and 0.625. Identical conditional code paths tied exactly. This confirms fiber fitting and restricted procedural separations. It does not fit an active law, train a full flow or diffusion, evaluate a complete VAE, measure full endpoint NLL, measure allocator memory or production timing, or supply observed-data evidence.

6 Discussion

The coherent conditional claim is structural. An invertible local chart retains the entire endpoint, a low-dimensional coarse state receives repeated computation, and a realizable stochastic fiber restores the complementary randomness. On the declared factorization this matches the exact endpoint and has latent-order local token work, while a non-Gaussian fiber strictly improves log score over a fixed Gaussian detail conditional. The comparator is not a complete variational autoencoder. A latent method that adopts the exact fiber ties, and natural data may violate local conditional closure. The next study therefore certifies which blocks can safely use one-shot fibers rather than assuming closure everywhere.

References

[1] Vikram Voleti, Chris Finlay, Adam Oberman, and Christopher Pal. Multi-resolution continuous normalizing flows. *arXiv:2106.08462*, 2021. `https://arxiv.org/abs/2106.08462`.

[2] Hao Phung, Quan Dao, and Anh Tran. Wavelet diffusion models are fast and scalable image generators. In *Proceedings of the IEEE/CVF Conference on Computer Vision and Pattern Recognition*, 2023. `https://openaccess.thecvf.com/content/CVPR2023/html/Phung_Wavelet_Diffusion_Models_Are_Fast_and_Scalable_Image_Generators_CVPR_2023_paper.html`.

[3] Andreas Blattmann et al. Align your latents: High-resolution video synthesis with latent diffusion models. In *Proceedings of the IEEE/CVF Conference on Computer Vision and Pattern Recognition*, 2023. `https://arxiv.org/abs/2304.08818`.